Quantitative Research Analyst

FF08 Data, analytics and AI · ISCO-08 2120 · CBS 0711 Biologen, wiskundigen en natuurwetenschappers · seniority medior · Also known as: Quantitative Researcher

Turns research questions into quantitative analyses, designs surveys or experiments and statistically tests whether observed relationships are reliable.

The quantitative research analyst collects and analyses numerical research data, from sample design to statistical testing of the outcomes. Unlike a data analyst who examines existing business data, they often design the research and the measurement approach themselves. Selection often looks at software skill, while recognising a bias in the sample before the conclusion is drawn makes the difference in practice between a defensible and a misleading outcome.

Key figures for this profile

25 skills · 3 competencies to measure · 6 knockout skills · centre of gravity: Digital, data and AI 64%, Cognitive ability and problem solving 18%, Communication and influence 11%

1 Competencies to measure

These are the behavioural and ability constructs in this profile. Each one shows which hrmforce questionnaire measures it.

Competencies to measure	Target level	hrmforce instrument	Assessment method	Behavioural anchor at target level
Numerical reasoning Knockout	N4	Ability Scan	Cognitieve capaciteitentest	Combines two data sources, calculates the development over time and explains which calculation step leads to which result.

Competencies to measure	Target level	hrmforce instrument	Assessment method	Behavioural anchor at target level
Curiosity and exploring Competency (50-framework): Leervermogen	N3	Big Fifty, 15PF, Motivatie	Persoonlijkheidsvragenlijst	Gathers information on new methods outside the own team and brings findings into the work meeting unprompted.
Dealing with ambiguity Competency (50-framework): Flexibiliteit	N3	Big Fifty, Mentale Weerbaarheid (4C), Competency Check	Situational Judgement Test	Formulates explicit assumptions when information is missing, proceeds on that basis and revises the approach once new data arrives.

2 Full skill profile

Every skill in this profile, sorted by weight. The behavioural anchor under each skill belongs to the target level.

T	Skill	Target level	Weight	Knockout	Assessment method	hrmforce instrument
A	Numerical reasoning Combines two data sources, calculates the development over time and explains which calculation step leads to which result.	N4	5	yes	Cognitieve capaciteitentest	Ability Scan
V	Critical thinking and source evaluation Detects fallacies and selective evidence in reports and states which conclusion the data do support.	N4	5	yes	Werkproef	Competency Check, Werkproef
V	Data analysis Selects analysis methods for an open question, turns raw data into a working file and supports every conclusion with figures.	N4	5	yes	Werkproef	Ability Scan, Werkproef
K	Basic statistical interpretation Interprets correlations, confidence intervals and significance in a report and points out where cause and effect are not established.	N4	5	yes	Kennistoets	Ability Scan, Kennistoets (klantspecifiek)

T	Skill	Target level	Weight	Knockout	Assessment method	hrmforce instrument
V	Visual communication and data storytelling Chooses a fitting chart type per message, removes unnecessary styling and writes the conclusion above the figure.	N4	4	no	Werkproef	Werkproef, Portfolio
V	Data visualisation Chooses the chart type per message, removes redundant elements and explains why an alternative display would mislead.	N4	4	no	Werkproef	Werkproef, Portfolio
V	Assessing data quality Defines validation rules for a data source, quantifies the defects and advises whether the data is fit for use.	N4	4	yes	Werkproef	Werkproef, Kennistoets (klantspecifiek)
V	Reporting and dashboard building Designs a dashboard together with the user, chooses indicators and definitions and processes user feedback into new versions.	N4	4	no	Werkproef	Werkproef, Portfolio

T	Skill	Target level	Weight	Knockout	Assessment method	hrmforce instrument
V	Testing hypotheses Turns a question into a testable hypothesis, chooses the test that fits the data and states the limitations of the conclusion.	N4	4	yes	Werkproef	Ability Scan, Werkproef
V	Data modelling Designs a logical data model for a defined domain, normalises where needed and discusses choices with users and developers.	N4	4	no	Werkproef	Kennistoets (klantspecifiek), Werkproef
V	Programming Independently builds a function or module from the requirement, chooses suitable structures and keeps the code readable and tested.	N3	4	no	Werkproef	Werkproef, Kennistoets (klantspecifiek)
K	Machine learning fundamentals Explains how a model is trained and validated and judges whether reported performance figures are credible.	N3	4	no	Kennistoets	Kennistoets (klantspecifiek)

T	Skill	Target level	Weight	Knockout	Assessment method	hrmforce instrument
V	Systems thinking Maps the chain around a bottleneck, points out the feedback loop and predicts the effect of an intervention.	N3	3	no	Werkproef	Competency Check, Werkproef
V	Methodical working Chooses a suitable method for a new task, records the steps and sticks to them under time pressure.	N3	3	no	Praktijkobservatie	Competency Check, Werkproef
V	Business reporting Builds a report from question to conclusion, separates fact from interpretation and phrases recommendations with a named owner.	N3	3	no	Werkproef	Werkproef, Competency Check
V	Advisory skill Reframes the question behind the question, offers two supported options with consequences and agrees a decision moment.	N3	3	no	Assessment center	Competency Check, Structureerd interview

T	Skill	Target level	Weight	Knockout	Assessment method	hrmforce instrument
G	Curiosity and exploring Gathers information on new methods outside the own team and brings findings into the work meeting unprompted.	N3	3	no	Persoonlijkheidsvragenlijst	Big Fifty, 15PF, Motivatie
V	Data management and storage Independently manages data collections, plans maintenance and recovery actions and tests whether a restore actually works.	N3	3	no	Werkproef	Kennistoets (klantspecifiek), Werkproef
V	Data governance and ownership Assigns ownership for a data domain, has definitions agreed and records decision making about data use traceably.	N3	3	no	Structureerd interview	Structureerd interview, Portfolio
K	GDPR and privacy in practice Judges in own processes whether a legal basis exists, applies data minimisation and handles requests from data subjects correctly.	N3	3	no	Kennistoets	Kennistoets (klantspecifiek)

T	Skill	Target level	Weight	Knockout	Assessment method	hrmforce instrument
V	Selecting and implementing AI applications Compares applications on requirements, risk and cost, runs a pilot and guides colleagues through adoption.	N3	3	no	Structureerd interview	Structureerd interview, Portfolio
V	AI governance and responsible use Tests intended AI use against policy and legislation, records the trade off and arranges human oversight of decisions.	N3	3	no	Kennistoets	Kennistoets (klantspecifiek), Structureerd interview
V	Recognising AI risk and bias Examines outcomes per group or situation, identifies where the system falls short and proposes measures or extra checks.	N3	3	no	Werkproef	Kennistoets (klantspecifiek), Werkproef
V	multidisciplinary collaboration Maps the issue with specialists from different disciplines and formulates a joint approach with a clear division of roles.	N3	2	no	Assessment center	360 Feedback, Competency Check

T	Skill	Target level	Weight	Knockout	Assessment method	hrmforce instrument
G	Dealing with ambiguity Formulates explicit assumptions when information is missing, proceeds on that basis and revises the approach once new data arrives.	N3	1	no	Situational Judgement Test	Big Fifty, Mentale Weerbaarheid (4C), Competency Check

3 Levels per seniority

The same profile with the job family seniority adjustment applied. Target levels are clamped to 1 to 5.

Skill	junior	medior	senior
Numerical reasoning	N3	N4	N5
Critical thinking and source evaluation	N4	N4	N4
Data analysis	N3	N4	N5
Basic statistical interpretation	N3	N4	N5
Visual communication and data storytelling	N3	N4	N5
Data visualisation	N3	N4	N5
Assessing data quality	N3	N4	N5
Reporting and dashboard building	N3	N4	N5
Testing hypotheses	N4	N4	N4
Data modelling	N3	N4	N5
Programming	N2	N3	N4
Machine learning fundamentals	N2	N3	N4
Systems thinking	N2	N3	N4
Methodical working	N2	N3	N4
Business reporting	N2	N3	N4
Advisory skill	N2	N3	N4
Curiosity and exploring	N2	N3	N4
Data management and storage	N2	N3	N4
Data governance and ownership	N2	N3	N4
GDPR and privacy in practice	N2	N3	N4
Selecting and implementing AI applications	N2	N3	N4
AI governance and responsible use	N2	N3	N4
Recognising AI risk and bias	N2	N3	N4
multidisciplinary collaboration	N2	N3	N4
Dealing with ambiguity	N2	N3	N4

4 The proficiency scale

Level	Label	Autonomy	Complexity	Scope	Depth of knowledge
N1	**Guided**	works under supervision and follows instruction	routine, one variable at a time	own task	basic understanding of terminology
N2	**Applying**	works independently within set frameworks	familiar situations with limited variation	own role	working knowledge, applies standard approach
N3	**Proficient**	sets own approach and seeks input proactively	several variables, some ambiguity	own team or process	thorough knowledge, weighs alternatives
N4	**Advanced**	directs others and decides on the approach	complex, with conflicting interests	several teams or a department	in-depth knowledge, advises others
N5	**Leading**	sets the standard and the policy	strategic, under high uncertainty	organisation, value chain or profession	authority, advances the profession

5 Behavioural anchors per skill

Anchors sit at N1, N3 and N5. N2 and N4 are deliberately not anchored: raters interpolate between them, following the O∗NET convention.

Skill	N1	N3	N5
Numerical reasoning Works with numbers, ratios and tables and draws conclusions from them about quantities, trends and relationships in numerical material.	Reads a simple table, names the highest and lowest value and calculates a percentage using a calculator.	Combines two data sources, calculates the development over time and explains which calculation step leads to which result.	Sees in complex number series which assumptions determine the outcome and sets the calculation logic that others in the organisation follow.
Critical thinking and source evaluation Tests claims for logic, evidence and the interest of the source and thereby distinguishes facts from opinions and assumptions.	Asks what a claim is based on and checks whether the source is named and can be found.	Detects fallacies and selective evidence in reports and states which conclusion the data do support.	Sets the assessment criteria for sources and claims, including for texts made by artificial intelligence, and enforces them organisation wide.
Data analysis Prepares and analyses data sets to find patterns, relationships and outliers and connects the outcomes to the original question.	Carries out a predefined analysis step on a supplied data set and records the outcome in the agreed format.	Selects analysis methods for an open question, turns raw data into a working file and supports every conclusion with figures.	Sets the organisation's analysis standards, reviews the analyses of others and establishes new analysis methods in the profession.

Skill	N1	N3	N5
Basic statistical interpretation Knows and interprets basic statistical concepts such as spread, correlation, confidence interval and significance and states what an outcome does and does not show.	Reads mean, spread and counts correctly from a table and checks what a percentage exactly represents.	Interprets correlations, confidence intervals and significance in a report and points out where cause and effect are not established.	Reviews the statistical basis of organisation wide decisions and trains others in the correct interpretation of results.
Visual communication and data storytelling Presenting figures and processes in charts, graphs and diagrams that support the message and lead the reader to the correct conclusion.	Creates a chart in an existing template, adds a title and names the unit shown on the axis.	Chooses a fitting chart type per message, removes unnecessary styling and writes the conclusion above the figure.	Sets guidelines for visualisation and dashboard use across the organisation and judges whether displayed figures represent reality fairly.
Data visualisation Turns data into charts and visual displays that show the message correctly, chooses suitable chart types and avoids misleading presentation.	Creates a chart following an example, completes axes and labels and checks whether the figures are correct.	Chooses the chart type per message, removes redundant elements and explains why an alternative display would mislead.	Sets the organisation's visualisation guidelines and judges whether reports by others represent the data honestly.
Assessing data quality Checks data sets for completeness, accuracy, duplicates and timeliness, describes the defects found and determines whether the data is usable.	Works through a checklist on a supplied data set and reports missing or duplicate records to the person responsible.	Defines validation rules for a data source, quantifies the defects and advises whether the data is fit for use.	Establishes the data quality framework for the entire organisation and decides which sources count as reliable.
Reporting and dashboard building Builds recurring reports and dashboards that show the right indicators for an agreed audience and keeps their definitions up to date.	Fills an existing dashboard or report template with current figures and checks the display for errors.	Designs a dashboard together with the user, chooses indicators and definitions and processes user feedback into new versions.	Determines which management information the organisation uses and keeps definitions unambiguous across all reporting.
Testing hypotheses Formulates a testable assumption, chooses a suitable test or comparison, carries it out and draws a conclusion with its uncertainty.	Carries out a provided test according to a step plan and records the outcome without adding own interpretation.	Turns a question into a testable hypothesis, chooses the test that fits the data and states the limitations of the conclusion.	Designs research set ups for the organisation and judges whether conclusions of others are supported by the data.

Skill	N1	N3	N5
Data modelling Describes data and its relationships in a model with entities, attributes and keys that matches the work process.	Extends an existing data model with new fields following the applicable naming conventions and has the work reviewed.	Designs a logical data model for a defined domain, normalises where needed and discusses choices with users and developers.	Sets the organisation's modelling standards and core data model and decides on deviations from them.
Programming Writes, reads and improves working program code in at least one language, splits a problem into steps and tests own solution.	Adjusts existing code as directed, runs it and reports error messages that cannot be resolved alone.	Independently builds a function or module from the requirement, chooses suitable structures and keeps the code readable and tested.	Sets programming standards and language choices for the organisation and reviews others' code for quality and maintainability.
Machine learning fundamentals Knows the main forms of machine learning, the role of training data and concepts such as overfitting, validation and model performance in plain language.	States the difference between supervised and unsupervised learning and which data a model needs to learn.	Explains how a model is trained and validated and judges whether reported performance figures are credible.	Sets requirements for model development in the organisation and reviews third party models on design and evidence.
Systems thinking Describes how parts of a process or organisation influence each other, including feedback, delay and unintended effects.	Names who or what depends on the own task earlier and later in the process.	Maps the chain around a bottleneck, points out the feedback loop and predicts the effect of an intervention.	Models the interplay between departments, market and regulation and shifts the intervention to the point with the greatest leverage.
Methodical working Follows a fixed and traceable method for tasks, with steps, documentation and checks, so that others can repeat the work.	Completes the steps of a prescribed procedure in the right order and records the result.	Chooses a suitable method for a new task, records the steps and sticks to them under time pressure.	Writes the working methods for the field, has them tested in practice and revises them at fixed moments.
Business reporting Bringing findings, figures and recommendations together in a report that leads the reader to a decision or next step in one reading.	Fills in an existing report template with the correct figures and describes per section what has been done.	Builds a report from question to conclusion, separates fact from interpretation and phrases recommendations with a named owner.	Sets the reporting line for the whole organisation, tests the quality of steering information and accounts for outcomes to board and regulator.

Skill	N1	N3	N5
Advisory skill Sharpening the question of a client and delivering a well founded recommendation in a way that the receiver adopts and implements it.	Supplies requested information and explains a standard recommendation following the fixed structure used by the team.	Reframes the question behind the question, offers two supported options with consequences and agrees a decision moment.	Advises the board on issues with conflicting interests, challenges established assumptions and safeguards the quality of advice across the organisation.
Curiosity and exploring Asks questions about the unfamiliar, seeks out new topics and perspectives unprompted and explores them further than the work strictly requires.	Asks follow up questions during an explanation about a new topic and reads the background material provided.	Gathers information on new methods outside the own team and brings findings into the work meeting unprompted.	Explores developments outside the sector systematically, connects unexpected findings and puts new themes on the agenda of the organisation.
Data management and storage Manages storage, access and lifecycle of data collections, applies changes in a controlled way and ensures backups and recovery work.	Carries out assigned maintenance tasks on a data collection and records every change in the log.	Independently manages data collections, plans maintenance and recovery actions and tests whether a restore actually works.	Determines how data storage is organised for the organisation and decides on platforms, retention methods and continuity requirements.
Data governance and ownership Records who owns and stewards which data, which definitions apply and how decisions about data use are made.	Maintains the register of data owners and definitions as instructed and reports ambiguities to the coordinator.	Assigns ownership for a data domain, has definitions agreed and records decision making about data use traceably.	Establishes the organisation's data governance model and holds executives to their role as owners of data.
GDPR and privacy in practice Knows privacy legislation requirements for own work, which legal basis and retention period apply and recognises when notification is required.	Processes personal data only according to the applicable work instruction and refers doubtful cases to the privacy officer.	Judges in own processes whether a legal basis exists, applies data minimisation and handles requests from data subjects correctly.	Sets privacy policy, decides on high risk processing operations and represents the organisation towards the supervisory authority.
Selecting and implementing AI applications Selects AI applications based on work need, cost, risk and fit with existing systems and guides adoption through to actual use.	Tests a designated AI application in own work and reports what proved usable and what did not.	Compares applications on requirements, risk and cost, runs a pilot and guides colleagues through adoption.	Determines which AI applications the organisation uses, decides on scaling up and accounts for the choices made.

Skill	N1	N3	N5
AI governance and responsible use Applies rules and agreements on permitted AI use, records where AI was used and ensures people remain accountable for decisions.	Uses AI only within permitted use, shares no confidential data and states where AI was used when this is required.	Tests intended AI use against policy and legislation, records the trade off and arranges human oversight of decisions.	Drafts the organisation's AI policy, sets up its oversight and accounts for it to regulators and clients.
Recognising AI risk and bias Recognises where an AI system works systematically skewed or unreliably for groups or situations and names consequences and possible measures.	States that AI output can be skewed by the data used and refers striking outcomes to a colleague.	Examines outcomes per group or situation, identifies where the system falls short and proposes measures or extra checks.	Sets the assessment frameworks for AI fairness and risk and decides whether a system may remain in use.
multidisciplinary collaboration Working with people from other disciplines by learning their language and methods and establishing shared starting points for the joint work.	Asks a colleague from another discipline what technical terms mean and summarises the answer back to them.	Maps the issue with specialists from different disciplines and formulates a joint approach with a clear division of roles.	Sets up lasting forms of collaboration between disciplines and settles conflicts between professional standards on the basis of the organisational interest.
Dealing with ambiguity Works on when the assignment, information or division of roles is incomplete or contradictory, makes workable assumptions and tests them once more becomes clear.	Asks what is meant when an assignment is unclear and then works according to the answer given.	Formulates explicit assumptions when information is missing, proceeds on that basis and revises the approach once new data arrives.	Sets direction under high uncertainty, makes the remaining uncertainty explicit and keeps the organisation moving until a decision can be taken.

6 Assessment methods and validity

The validities are the corrected meta-analytic estimates from Sackett, Zhang, Berry and Lievens (2022) in the Journal of Applied Psychology, which replace the older Schmidt and Hunter (1998) figures. The spread matters as much as the mean: with a high SD, validity varies strongly by context.

Assessment method	rho	SD	What it is
Cognitieve capaciteitentest	0.41	0.14	Standardised test of verbal, numerical or abstract reasoning.
Werkproef	0.33	0.09	The candidate performs a representative work sample under standardised conditions.
Kennistoets	0.40	0.13	Test of declarative job knowledge, usually assembled per client.
Praktijkobservatie	-	-	Structured workplace observation with a form based on behavioural anchors.
Assessment center	0.29	0.09	Multiple exercises, multiple assessors, dimension-wise scoring.
Persoonlijkheidsvragenlijst	0.40	0.15	Self-report of behavioural preferences. At hrmforce the Big Fifty and the 15PF.
Structureerd interview	0.42	0.19	Behaviour-based interview with fixed questions and a rating scale per question.
Situational Judgement Test	0.12	0.11	Scenario with response options, scored on knowledge or behavioural tendency.

7 Assessment instruments for this job

hrmforce instrument	Skills	What it measures
Werkproef	14	Representatief werkfragment onder gestandaardiseerde condities
Kennistoets (klantspecifiek)	9	Toets van vakkennis, per klant samengesteld
Competency Check	7	Competentiebeoordeling op gedragsniveau
Portfolio	5	Beoordeling van eerder werk tegen vaste criteria
Ability Scan	4	Capaciteitentest: verbaal, numeriek en abstract redeneervermogen
Structureerd interview	4	Gedragsgericht interview met vaste opzet
Big Fifty	2	Persoonlijkheidsvragenlijst, 150 items, 25 factoren
15PF	1	Persoonlijkheidsprofiel, korter alternatief voor de Big Fifty

hrmforce instrument	Skills	What it measures
Motivatie	1	Drijfveren en motivatiebronnen
360 Feedback	1	Meervoudige beoordeling op gedragsankers
Mentale Weerbaarheid (4C)	1	Mentale weerbaarheid volgens het 4C-model

8 How to use this profile

1. Agree the profile with the hiring manager before the first candidate is measured. Weights shifted afterwards make the outcome indefensible.
2. Choose the assessment method per skill from the Assessment method column. Combine at most three methods per candidate.
3. Calibrate the assessors in a two-hour session using the anchors in chapter 5. Anchors without rater training deliver little.
4. Fill in the match calculator and discuss the gaps in the development conversation. Type A and type G are selection criteria, not one-quarter development goals.

Source: hrmforce Skills Framework 2.0, version 2.0.0, built 2026-09-08. Crosswalks to ESCO v1.2.1, O*NET 31.0, Lightcast Open Skills, SFIA 9, DigComp 3.0, EntreComp, LifeComp, GreenComp, AILit, WEF Future of Jobs 2025, ISCO-08 and the Dutch CBS occupational classification BRC 2014 edition 2025. This profile is a starting point, not a standard: align it with your own organisation before selecting on it.

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